

1. Employee Salary Analysis

A CSV file (employees.csv) contains employee details.

Task

- ❑ Read the CSV file.
- ❑ Display employees earning more than ₹50,000.
- ❑ Calculate average salary department-wise.

```
import pandas as pd

df = pd.read_csv(r"c:\Users\hp\Downloads\employee.csv")

print(df)

print(df[df["Salary"] > 50000])

print(df.groupby("Department")["Salary"].mean())
```

✓ 4.7s

	EmpID	Name	Department	Salary
0	101	Ravi	HR	45000
1	102	Priya	IT	60000
2	103	Rahul	IT	55000
3	104	Anita	Sales	40000
4	105	Kiran	HR	70000

	EmpID	Name	Department	Salary
1	102	Priya	IT	60000
2	103	Rahul	IT	55000
4	105	Kiran	HR	70000

Department

HR	57500.0
IT	57500.0
Sales	40000.0

Name: Salary, dtype: float64

2. Missing Data Handling

Given a CSV file containing student marks with missing values.

Task

- ❑ Identify missing values.
- ❑ Replace missing marks with the column average.
- ❑ Display the cleaned dataset.

```
import pandas as pd

df = pd.read_csv(r"C:\Users\hp\Downloads\Student.csv")

print(df.isnull().sum())

df.fillna(df.mean(numeric_only=True), inplace=True)

print(df)
```

[4] ✓ 0.0s

...

	RollNo	Name	Marks
0	1	Ravi	80.000000
1	2	Priya	81.666667
2	3	Rahul	75.000000
3	4	Anita	90.000000
4	5	Kiran	81.666667

3. Duplicate Record Removal

A customer dataset contains duplicate rows.

Task

- 🔗 Read the dataset.
- 🔗 Find duplicate records.
- 🔗 Remove duplicates.
- 🔗 Display the updated dataset.

```
import pandas as pd

df = pd.read_csv(r"C:\Users\hp\Downloads\Customer.csv")

print(df[df.duplicated()])

df = df.drop_duplicates()

print(df)
```

✓ 0.0s

	CustomerID	Name	City
3	2	Priya	Chennai
	CustomerID	Name	City
0	1	Ravi	Hyderabad
1	2	Priya	Chennai
2	3	Rahul	Bangalore
4	4	Anita	Delhi

8. Data Aggregation

Given monthly sales data.

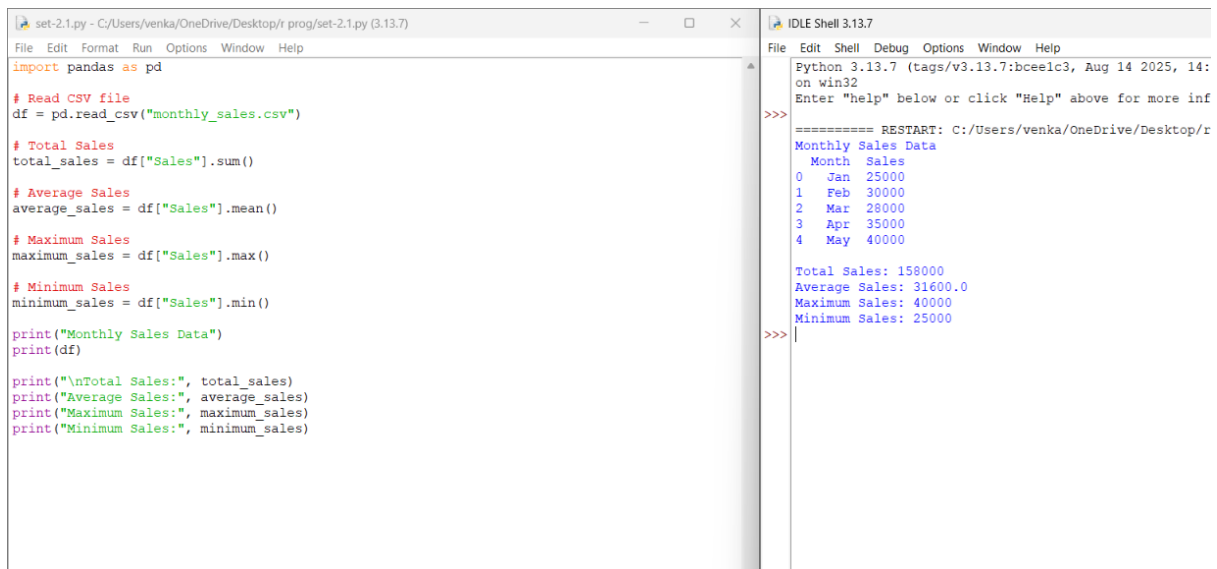
Task

❑ Calculate total sales.

❑ Average sales.

❑ Maximum sales.

❑ Minimum sales.



The image shows a Python script in a text editor window titled 'set-2.1.py' and its execution output in the IDLE Shell 3.13.7.

Python Script (set-2.1.py):

```
import pandas as pd

# Read CSV file
df = pd.read_csv("monthly_sales.csv")

# Total Sales
total_sales = df["Sales"].sum()

# Average Sales
average_sales = df["Sales"].mean()

# Maximum Sales
maximum_sales = df["Sales"].max()

# Minimum Sales
minimum_sales = df["Sales"].min()

print("Monthly Sales Data")
print(df)

print("\nTotal Sales:", total_sales)
print("Average Sales:", average_sales)
print("Maximum Sales:", maximum_sales)
print("Minimum Sales:", minimum_sales)
```

IDLE Shell Output:

```
Python 3.13.7 (tags/v3.13.7:bceelc3, Aug 14 2025, 14:
on win32
Enter "help" below or click "Help" above for more inf
>>>
===== RESTART: C:/Users/venka/OneDrive/Desktop/r
Monthly Sales Data
  Month Sales
0   Jan  25000
1   Feb  30000
2   Mar  28000
3   Apr  35000
4   May  40000

Total Sales: 158000
Average Sales: 31600.0
Maximum Sales: 40000
Minimum Sales: 25000
>>>
```

9. Grouping Data

Given employee information.

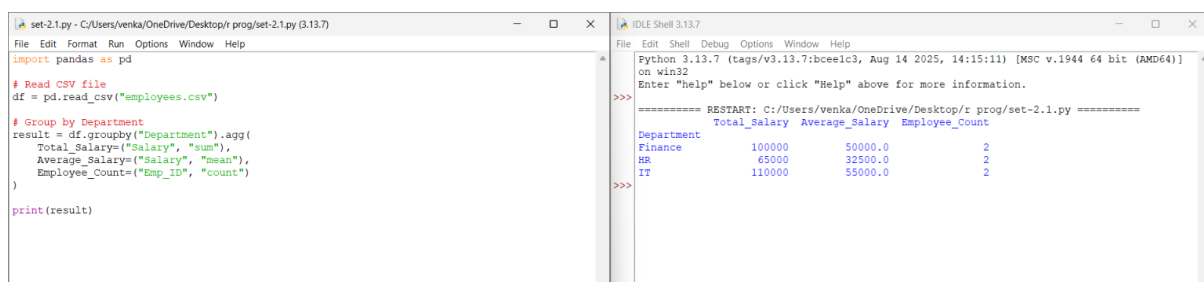
Task

Group employees by department and calculate:

❓ Total salary

❓ Average salary

❓ Employee count



```
set-2.1.py - C:/Users/venka/OneDrive/Desktop/r prog/set-2.1.py (3.13.7)
File Edit Format Run Options Window Help
import pandas as pd

# Read CSV file
df = pd.read_csv("employees.csv")

# Group by Department
result = df.groupby("Department").agg(
    Total_Salary=("Salary", "sum"),
    Average_Salary=("Salary", "mean"),
    Employee_Count=("Emp_ID", "count")
)

print(result)
```

```
IDLE Shell 3.13.7
Python 3.13.7 (tags/v3.13.7:bce1c3, Aug 14 2025, 14:15:11) [MSC v.1944 64 bit (AMD64)]
on win32
Enter "help" below or click "Help" above for more information.
>>>
===== RESTART: C:/Users/venka/OneDrive/Desktop/r prog/set-2.1.py =====
      Department  Total_Salary  Average_Salary  Employee_Count
Finance          100000.0         50000.0             2
HR                65000.0         32500.0             2
IT                110000.0         55000.0             2
>>>
```

10. Merge Two CSV Files

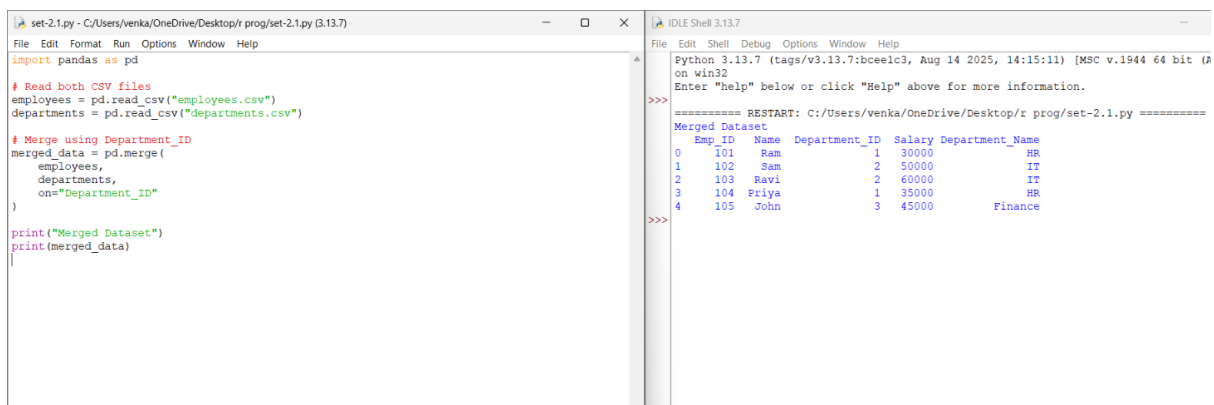
Files:

employees.csv

departments.csv

Task

Merge the two datasets using Department ID.



The screenshot shows a Python IDE with two windows. The left window, titled 'set-2.1.py', contains the following code:

```
import pandas as pd

# Read both CSV files
employees = pd.read_csv("employees.csv")
departments = pd.read_csv("departments.csv")

# Merge using Department_ID
merged_data = pd.merge(
    employees,
    departments,
    on="Department_ID"
)

print("Merged Dataset")
print(merged_data)
```

The right window, titled 'IDLE Shell 3.13.7', shows the output of the script. It displays a restart message and then a table representing the merged dataset:

```
===== RESTART: C:/Users/venka/OneDrive/Desktop/r prog/set-2.1.py =====
Merged Dataset
  Emp_ID  Name  Department_ID  Salary  Department_Name
0    101   Ram              1   30000                HR
1    102   Sam              2   50000                IT
2    103   Ravi              2   60000                IT
3    104  Priya              1   35000                HR
4    105   John              3   45000              Finance
```

8. Data Aggregation

Given monthly sales data.

Task

- ☐ Calculate total sales.
- ☐ Average sales.
- ☐ Maximum sales.
- ☐ Minimum sales.

code:

```
import pandas as pd

data = {
    "Month": ["Jan", "Feb", "Mar", "Apr", "May"],
    "Sales": [25000, 30000, 28000, 35000, 40000]
}

df = pd.DataFrame(data)

print("Sales Data")
print(df)

print("\nTotal Sales:", df["Sales"].sum())
print("Average Sales:", df["Sales"].mean())
print("Maximum Sales:", df["Sales"].max())
print("Minimum Sales:", df["Sales"].min())
|
```

Output:

```
----- RESIARI. D./query/CO1 act.py -----
Sales Data
  Month  Sales
0   Jan  25000
1   Feb  30000
2   Mar  28000
3   Apr  35000
4   May  40000

Total Sales: 158000
Average Sales: 31600.0
Maximum Sales: 40000
Minimum Sales: 25000
|
```

10. Merge Two CSV Files

Files:

- ☐ employees.csv
- ☐ departments.csv

Task

Merge the two datasets using Department ID.

Code:

```

import pandas as pd

employees = {
    "Emp_ID": [101, 102, 103, 104],
    "Name": ["Rahul", "Priya", "Amit", "Sneha"],
    "Dept_ID": [1, 2, 1, 3]
}

departments = {
    "Dept_ID": [1, 2, 3],
    "Department": ["IT", "HR", "Finance"]
}

emp_df = pd.DataFrame(employees)
dept_df = pd.DataFrame(departments)

merged_df = pd.merge(emp_df, dept_df, on="Dept_ID")

print("Merged Data")
print(merged_df)

```

Output:

```

===== RESTART: D:/query/col atl 8th quetion.py =====
Merged Data
   Emp_ID  Name  Dept_ID Department
0     101  Rahul        1          IT
1     102  Priya        2          HR
2     103   Amit        1          IT
3     104  Sneha        3      Finance

```

17. Data Interpretation using Python

Given marks of students.

Task

☐ Store marks in a list.

☐ Find:

o Highest mark

o Lowest mark

o Average mark

o Number of students passed.

Code:

```
marks = [75, 82, 65, 48, 90, 55]

print("Student Marks:", marks)

print("Highest Mark:", max(marks))
print("Lowest Mark:", min(marks))
print("Average Mark:", sum(marks) / len(marks))

passed = 0

for mark in marks:
    if mark >= 50:
        passed += 1

print("Students Passed:", passed)
```

Output:

```
===== RESTART: D:/query/col at1 17 question.py =====
Student Marks: [75, 82, 65, 48, 90, 55]
Highest Mark: 90
Lowest Mark: 48
Average Mark: 69.16666666666667
Students Passed: 5
```